

TAI, Wing-Kuen (William)

Phone: (+852) 5928 8549 | Email: williamtaiwk@gmail.com | GitHub: github.com/WilliamTai0 | [LinkedIn](#)

Summary

Final-year Computer Engineering student at HKUST with hands-on full-stack development experience across AI-powered products — from backend APIs (FastAPI, REST, MongoDB/MySQL) to frontend testing (Vitest, React Testing Library) and mobile app development. Proven ability to integrate AI agents and LLM APIs into production workflows, ship features end-to-end, and move fast in CI/CD-driven environments (GitHub Actions, Docker). High-agency builder with a track record of owning projects from concept to Grade-A delivery.

Technical Skills

AI / AI-Native Workflow: LLM API Integration, Function Calling, Prompt Engineering, Model Context Protocol (MCP), AI Agent Tooling

Full-Stack: FastAPI, REST API Design, JavaScript, User Authentication, CRUD Architecture (MVVM)

Testing & DevOps: Vitest, React Testing Library, MSW, GitHub Actions (CI/CD), Docker, Docker Compose, Branch Protection Rules

Databases: MongoDB Atlas, GridFS, MySQL

Programming Languages: Python, JavaScript, C++, Java, C

Tools: Git/GitHub, VS Code, Antigravity, Unity

Projects

AI Developer Assistant Integration

Personal Project | Jan 2026 – Present

- Built and deployed AI agent tool servers via Model Context Protocol (MCP), sandboxed in Docker, to extend LLM runtime context — applying an AI-native workflow to accelerate development velocity.

[Smart Trip Planner](#)

COMP4521 Mobile App Dev | Android Developer | Sep – Nov 2025

- Shipped an AI-powered travel app end-to-end in Flutter, integrating Gemini 2.5 Flash Lite with function-calling to ground LLM output in real-time Amadeus and Google Places API data.
- Owned authentication, MongoDB Atlas backend, and CRUD trip management (MVVM/Provider), plus home screen UI/navigation and end-to-end Android testing.

[AI-Powered Health App](#)

HKUST FYP | Backend & Database Developer | Jun 2025 – May 2026

- Architected an end-to-end backend (FastAPI, Uvicorn, Linux) serving 20+ RESTful endpoints for auth, leaderboards, and social features; managed media via MongoDB GridFS. Awarded Grade A.
- Integrated Qwen-2.5-VL via Hugging Face API with strict JSON schema enforcement to eliminate parser instability.
- Containerized a multi-model ML pipeline (YOLOv8, PaddleOCR, OpenVINO VLM) with Docker Compose, resolving cross-library conflicts for 150MB+ model weights.

Education

HKUST | BEng in Computer Engineering | Sep 2023 – May 2026

HKU SPACE Community College | Associate of Engineering in Computer Science | Sep 2021 – Aug 2023

Certifications

[Gemini Certified University Student](#) Google for Education | Jun 2026

[AI Code Generation for Everyone](#) HKUST via Coursera | Jun 2026

[Search Engines for Web and Enterprise Data](#) HKUST via Coursera | May 2026

Training

Linux System & Network Administration HKUST Industrial Training | Jan 2026

- Automated system tasks via Shell scripting; administered firewall/network policies (nftables/firewalld) in VMware.

Cloud Services in Azure HKUST Industrial Training | Jan 2026

- Provisioned Azure SQL Database, configured Key Vault secrets, evaluated cloud cost via TCO Calculator.

Languages

Cantonese (Native) | English (Fluent) | Mandarin (Fluent)